

DSN Mart Sales Prediction

Model Results & Business Communication Report

1. Executive Summary

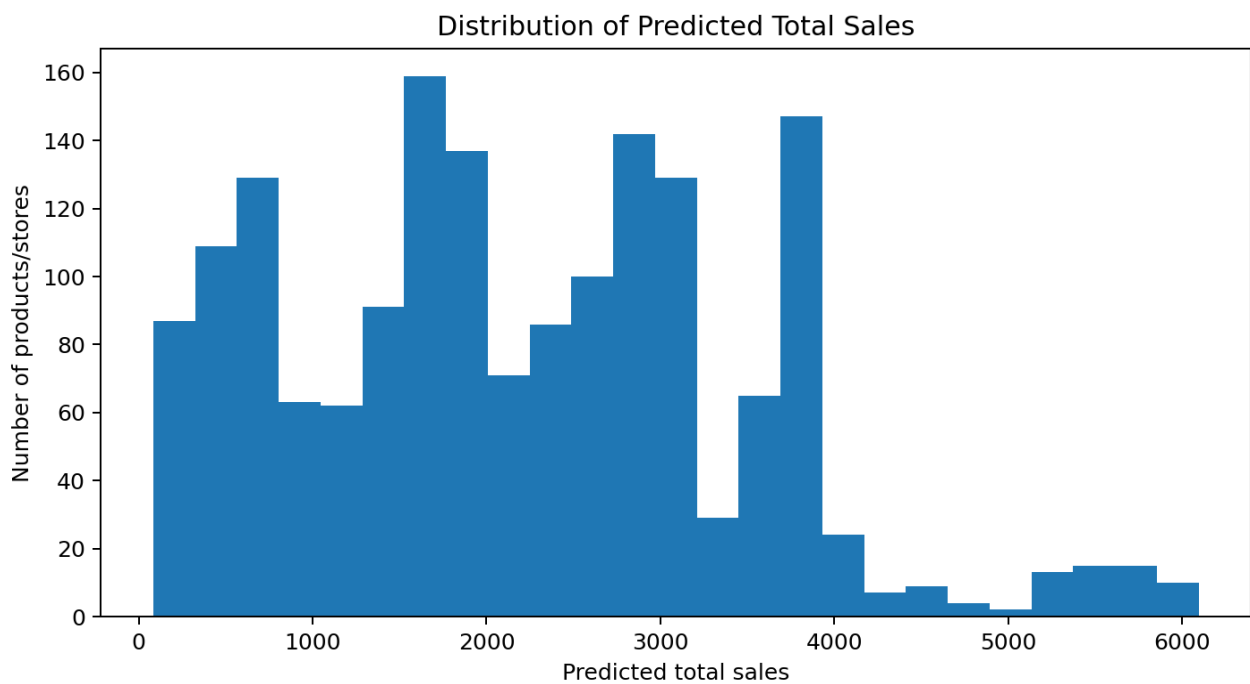
The final submission contains **1,705 product-store predictions** for total sales. The model's predicted sales range from **83.03** to **6,095.49**, with an average prediction of **2,192.61** and a median of **2,021.07**.

The output is suitable as a forecasting layer for comparing expected sales across product-store combinations. The submission contains one prediction per unique ID and has no missing prediction values.

2. Final Prediction Results

Metric	Result
Number of predictions	1,705
Average predicted sales	2,192.61
Median predicted sales	2,021.07
Minimum predicted sales	83.03
Maximum predicted sales	6,095.49
Standard deviation	1,287.62
25th percentile	1,211.47
75th percentile	3,061.33

3. Prediction Distribution



The distribution shows that predicted sales are spread across a relatively broad range rather than being concentrated at a single value. The middle 50% of predictions lie between 1,211.47 and 3,061.33.

4. Modelling Approach

The modelling workflow combines the training and test datasets during consistent cleaning, while retaining a flag to separate training rows from test rows. Text categories are standardised by trimming whitespace and normalising case.

Missing product weight is imputed first using the median weight for the same product and then using the product-category median as a fallback. Missing store size is retained as an "Unknown" category rather than being assigned an arbitrary store size.

Two additional features are engineered: **price_per_kg** and **store_profile**, which combines store format and store size. Numeric features are passed through, store size and location tier are ordinal-encoded, and nominal categories are one-hot encoded.

The final estimator is a **Random Forest Regressor** with 600 trees, maximum depth of 6, minimum leaf size of 8, and `max_features` set to 0.6. The model was designed to predict continuous total sales values.

5. Validation and Model Assessment

The modelling script uses 5-fold cross-validation and RMSE (Root Mean Squared Error) as the validation metric. The supplied final submission file contains predictions only; it does not contain the observed target values or the printed cross-validation RMSE. Therefore, a numerical model-accuracy claim cannot be made from the final submission alone.

This distinction is important when communicating the result: the figures in this report describe the model's predictions, not measured prediction accuracy against unseen actual sales.

6. Business Interpretation

The predictions provide an expected-sales estimate for each product-store combination. In a business setting, these estimates can support prioritisation of inventory planning, store-level product reviews, demand monitoring, and identification of combinations with comparatively high or low expected sales.

The average prediction of 2,192.61 provides a general reference point, while the range from 83.03 to 6,095.49 indicates substantial variation between product-store combinations. The median of 2,021.07 is below the mean, indicating that higher predicted values pull the average upward.

7. Key Takeaways

- The final submission contains 1,705 complete sales predictions.
- Predicted sales range from 83.03 to 6,095.49.
- Average predicted sales are 2,192.61; median predicted sales are 2,021.07.
- The workflow includes data cleaning, missing-value treatment, feature engineering, categorical encoding, and Random Forest regression.
- 5-fold cross-validation with RMSE is part of the modelling workflow, but its numerical result is not present in the supplied submission file.
- The predictions can be used as an input to inventory and demand-planning analysis, subject to validation against actual sales.

8. Recommended Next Step

For a complete model-performance report, add the cross-validated RMSE from the training run and, if actual test outcomes later become available, compare predictions with actual sales using RMSE, MAE, and an appropriate error analysis. The feature-importance output from the training run can also be included to explain which variables contributed most strongly to the model's predictions.